

IAT 461 Assignment 3 Report

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A1

Cleaning decisions

- Cleaning 1 - status filter. Keep only Issued rows because these only represent businesses that are active.
- Cleaning 2 - geom column. Remove all of geom column because it is 100% missing on every row. Nothing todo in the code. I am ignoring this column.
- Cleaning 3 - missing coordinates. Half of rows have no location (mostly things like Short-term Rental Operators with no fixed storefront). Rather than dropping them, keep them and flag with has_geo, only dropping when a step actually needs lat/lon.
- Cleaning 4 - businesssubtype. Most are missing, too sparse to be useful. This does not add much more categorisation. business type is more useful. Nothing todo in the code. I am ignoring this column.
- Cleaning 5 - postcode. Half are missing, too incomplete. Use localarea instead because only a few area missing.
- Cleaning 6 - feepaid imputation. Some are missing. Fill with the median fee for that business's business type, falling back to the overall median if a type has no observed fees at all.
- Cleaning 7 - business type consolidation. 94 long-tailed categories. Keep the top 20 (covers most of the records) and bucket everything else as "Other".

A2

K-means vs. DBSCAN

1. K-means (K=6) maps cleanly onto real Vancouver neighbourhoods. Each cluster is dominated by a recognizable area. K-means splits the city into 6 compact, contiguous areas that match how a local would describe Vancouver's areas.

2. DBSCAN (eps=0.2, min_samples=25): there is one giant cluster that includes Downtown. Business density there is high and uniform with no real gaps. The other 16 small clusters sit in lower density outer neighbourhoods that do have gaps: Sunset, Marpole, Oakridge, Dunbar-Southlands, Shaughnessy/Kerrisdale, West Point Grey, Killarney, Victoria-Fraserview, Hastings-Sunrise, Arbutus-Ridge.

Why they differ. K-means always carves the map into K similarly sized regions regardless of density, so it slices the uniform urban core into separate areas. DBSCAN only splits where there's an actual density gap, so it keeps the core as one cluster but correctly separates the sparser outer neighbourhoods. Both methods agree those outer areas are distinct. DBSCAN just carves them out explicitly instead of forcing them into a nearby area.

DBSCAN's noise are isolated businesses in lower density areas with too few nearby neighbours to join a cluster. K-means has no "noise" option. It would force these into the closest area even when that's a poor fit.

A3

Size/Industry/Lifecycle clusters

- Cluster 0 - Health Care, Retail Dealer, Beauty Services
- Cluster 1 - Business Support Services
- Cluster 2 - Long Term Rental
- Cluster 3 - Other
- Cluster 4 - Legal Services
- Cluster 5 - Association or Society

Does this differ from A2's geographic clustering? Yes. It is very different from A2 because there does not seem to be a correlation between the location and the type of business. Every feature cluster is spread evenly across all six geographic clusters. Where a business is located shows almost nothing about what kind of business it is, its size, or its lifecycle.

B4

At $K=4$, this grouping is genuinely meaningful, and it does cut across geography.** One cluster of 14 areas spans opposite ends of the city: Kerrisdale, Dunbar-Southlands, West Point Grey, Arbutus-Ridge on the west side sit in the *same* cluster as Renfrew-Collingwood, Killarney, Victoria-Fraserview, Hastings-Sunrise on the east side, plus Kensington-Cedar Cottage, Riley Park, Oakridge, Shaughnessy, South Cambie, and Kitsilano. Every one in those areas is dominated by long term rental licences, because they are all primarily residential neighbourhoods rather than commercial districts. K-means is correctly picking up residential character neighbourhood as a real pattern that has nothing to do with which side of the city you are on.

Downtown, Marpole, Mount Pleasant, Sunset, Grandview-Woodland, West End** form a mixed commercial/dense cluster, while **Fairview** and **Strathcona** are distinctive enough to stand alone as singleton clusters at this K value.

This lines up with the same conclusion from Part A: the location based K-means/DBSCAN clusters and the Size/Industry/Lifecycle feature clusters on individual businesses had essentially zero agreement. The neighbourhood level result geography and what kind of business mix an area has are close to independent signals in this dataset.

B5

- Do areas of similar businesses also tend to be geographically close? No, because at $K=4$ the largest cluster contains west side areas and east side areas together, driven by a shared long term rental dominated business mix rather than proximity.
- How does this compare to Part A's Industry clustering? The same near zero Adjusted Rand Index between geographic and feature based clusters shows up at both the individual business level and here at the neighbourhood level.
- One unexpected pairing: Dunbar-Southlands and Renfrew-Collingwood land in the same cluster at $K=4$ despite being on opposite sides of the city and having very different reputations. What they share is a business landscape that is overwhelmingly long term rental plus general contractor. For example, both are primarily residential areas with light home services activity rather than commercial hubs.